

FlexBox firmware — Backend uplink/downlink cheatsheet

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Uplink / downlink schema (reference for backend)

Payload layout, FPorts, event bytes, and application downlinks. Device is **LoRaWAN Class A, OTAA**; application payload length is **11 bytes** (PAYLOAD_LEN_BYTES / DR0-style fixed frame).

All uplink timestamps in this document are **UTC epoch seconds, big-endian**, bytes **0–3**, unless noted otherwise.

Firmware version this doc was aligned with: see `v1.1/app/include/sys_config.h` (FW_VERSION_* / HW_VERSION_*). Regenerate or bump the date when you

change payloads.

FPorts (uplink routing)

FPort	Use
10	Button (public vote), event 0x00
11	NFC (check-in 0x01 , check-out 0x02 , registered vote 0x03)
20	Housekeeping: battery 0x10 , counter sync 0x12 , device state snapshot 0x13
21	Device info: firmware/hardware version 0x14 (queued only after downlink 0x08)
30	Alarms (reserved; not used on air yet)
13	Reserved macro FPORT_FUTURE in firmware — no defined uplinks today

Downlink: Application commands are selected by **first FRMPayload byte** (command). The dispatcher logs **port** but **does not route** on FPort; any application FPort is accepted for these commands.

Uplink confirmation policy (important for RX windows)

Configured in `sys_config.h` (field-stable profile):

Traffic	Typical confirmed	Notes
Button 0x00	Unconfirmed	Avoids RX timeout on weak links
NFC 0x01–0x03	Confirmed	ACK expected; use for provisioning / staff flows
Housekeeping 0x10, 0x12, 0x13	Unconfirmed	Includes periodic heartbeat battery frame
Device info 0x14	Unconfirmed	Sent after DL 0x08

Class A rule: downlink to the device is only possible in **RX1/RX2** following an **uplink** the network accepted. Prefer **confirmed** NFC when you must deliver a downlink in the next receive slot.

Uplink: Event codes and payloads (11 bytes)

Common: [0..3] = timestamp (uint32 BE), [4] = event code. Remaining bytes are event-specific.

0x00 — Button (public vote)

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x00
button_id	5	0..5
counter	6–8	24-bit BE (per-button)
reserved	9–10	0

- **FPort:** 10
 - **When:** User presses a feedback button.
 - **Backend:** Use counter for idempotency / reconciliation.
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0x01 — NFC check-in

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x01
uid	5–8	4-byte card UID
reserved	9–10	0

- **FPort:** 11
 - **When:** Staff NFC check-in (staff flow / button 0 context in UI).
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0x02 — NFC check-out

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x02
uid	5–8	4-byte card UID
reserved	9–10	0

- **FPort:** 11
- **When:** Staff NFC check-out. Device updates “Last Cleaned” locally for display.

0x03 — NFC registered vote

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x03
button_id	5	0..5
card_data	6–9	4-byte card UID
reserved	10	0

- **FPort:** 11
- **When:** Staff taps NFC then selects a button (registered vote).

0x10 — Battery status (heartbeat)

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x10
battery_mv	5–6	uint16 BE (mV)
percent	7	0..100 (currently 0 in housekeeping path; reserved)
flags	8	bitfield (0 for now)
reserved	9–10	0

- **FPort:** 20
- **When:** Part of **housekeeping** (**housekeeping_run**): periodic daily tick (with DevEUI jitter) and after downlink **0x04** status request.

- **Backend:** Treat as heartbeat / telemetry; good moment to queue a downlink for the next uplink.

Heartbeat timing (jitter): When enabled, first run each UTC day at:

`heartbeat_time_UTC = 00:00 UTC + offset_minutes`

`offset_minutes = (DevEUI[7]*256 + DevEUI[6]) % 1440`

(DevEUI is 8 bytes; indices 6 and 7 are the last two octets.)

0x12 — Counter sync

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x12
button_id	5	0..5
counter	6–8	24-bit BE
reserved	9–10	0

- **FPort:** 20
- **When:** After join and during housekeeping; rate-limited / staggered across buttons.
- **Backend:** Reconcile per-device, per-button counters.

0x13 — Device state snapshot (new since 1.2.1 cheatsheet)

Field	Bytes	Format
ts	0–3	uint32 BE (uplink time; RTC or monotonic fallback)
evt	4	0x13
last_cleaned_epoch	5–8	uint32 BE (epoch seconds for EPD “Last Cleaned”; 0 if unset)
tz_offset_minutes	9–10	int16 BE (signed minutes to add to UTC for local display; same encoding as DL 0x03 , negative = west of UTC)

- **FPort:** 20

- **When:** Queued after join-related flows, with housekeeping, after DL **0x03** (timezone write succeeds), and as part of the **0x04** status-request path.
- **Backend:** Single frame for “what the device thinks” about last cleaned + timezone.

0x14 — Firmware / hardware version + EPD line

11 bytes. This event **does not** use the usual leading UTC timestamp ([0..3] + [4]=evt). Decoders must branch on **FPort 21** and/or `payload[0]==0x14`.

Field	Bytes	Format
evt	0	0x14
fw_major	1	uint8
fw_minor	2	uint8
fw_patch	3	uint8
hw_major	4	uint8
hw_minor	5	uint8
hw_patch	6	uint8
epd_enabled	7	0 = no EPD in build (variant A), 1 = EPD in build (variant B). Mirrors preprocessor EPD_ENABLED in sys_config.h .
reserved	8–10	0

- **FPort: 21**
- **When:** Only after application downlink **0x08** (version query).
- **Backend:** Read semver from bytes **1–6**; byte **7** selects UI / inventory bucket for display vs headless builds.

0x11 — Low battery alarm

- Defined in firmware; **not** asserted on the wire in current builds. Same FPort family as housekeeping when implemented.

0xFF — Reserved

- Reserved for future event types.

Downlink: Application commands (FRMPayload)

First byte = **command**. Length must satisfy each command's minimum (see below).

Cmd	Code	Min len	Payload shape	Effect
EPD “Last Cleaned” epoch	0x01	5	01 + epoch uint32 BE	Store / display “Last Cleaned” on EPD from epoch.
EPD full refresh	0x02	1	02	Full e-paper refresh (reduce ghosting).
Timezone offset	0x03	3	03 + int16 BE signed minutes (add to UTC for local civil time)	Persist TZ for local “Last Cleaned”; refresh display; queue 0x13 snapshot.
Status / house- keeping	0x04	1	04	Runs house- keeping path: battery 0x10 (if ADC OK), LinkCheck- Req , counter-sync burst, 0x13 snapshot, schedules time sync after burst — not “battery only” anymore.
Reset counters	0x05	1	05	Reset all button counters (EEPROM); uses power rail for store access.

Cmd	Code	Min len	Payload shape	Effect
Factory reset	0x06	1	06	Counters + DevNonce + has_joined_once cleared; LED pattern then delayed reboot (first-boot style join).
Reboot	0x07	1	07	Request controlled reboot via SMF.
FW/HW query	0x08	1	08	Queue uplink 0x14 on FPort 21 (compact layout: no ts; semver + EPD flag).
Custom EPD text	0x99	2	99 + dur_min + body	Body = ASCII hex pairs (even count); decoded bytes must be printable 0x20–0x7E . Shows fullscreen banner for dur_min minutes; dur_min = 0 uses <code>DL_CUSTOM_TEXT_DEFAULT_MINUTES</code> from <code>sys_config.h</code> . Then reverts to normal cleaning / last-cleaned UI.

Unknown first-byte commands are logged and ignored.

Timezone examples (0x03 payload = 03 + int16 BE signed offset minutes):

Region	Offset (min)	Hex payload
New York (EST, winter)	−300	03 FE D4
New York (EDT, summer)	−240	03 FF 10
Denver / Mountain (MST)	−420	03 FE 5C
Denver / Mountain (MDT)	−360	03 FE 98
San Francisco / Pacific (PST)	−480	03 FE 20
San Francisco / Pacific (PDT)	−420	03 FE 5C

To switch a device: send one downlink with the 3-byte payload above. No reboot required for TZ alone.

How to send: Queue application downlink with FRMPayload = command byte + arguments. FPort is not interpreted for command selection (any app port, e.g. 10, is fine). **Class A:** expect delivery in RX1/RX2 after the next accepted uplink.

EPD update example: Epoch 1771131600 → 01 69 91 51 90. <https://www.epochconverter.com/hex>

Custom text example (0x99): Message "Hi" → ASCII 48 69 → hex digits 4869 as ASCII in payload: 99 05 34 38 36 39 = cmd 99, dur 5 min, hex body 4869 (pairs: 48→‘H’, 69→‘i’).

Notes

- **0x04** is the right command when you want a **telemetry burst** and an immediate chance for **downlink** on the subsequent uplinks (battery + counters + **0x13** snapshot, then coordinated time sync).
 - Button IDs **0..5** (six buttons). Counters are **24-bit BE** in button and counter-sync frames.
 - **0x13** and **0x14** are the main **backend-visible** additions since the 2026-02-24 1.2.1 cheatsheet; downlink **0x07**, **0x08**, and **0x99** are new commands.
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Example payloads (reference hex)

Decoded samples. Bytes **0–3** = UTC epoch (uint32 BE); byte **4** = event code.

FPort	Base64	Hex	Explanation
10	aZ3guwAAAAAEAAA=	69 9d e0 bb 00 00 00 00 04 00 00	Button (0x00): ts 0x699DE0BB, button 0, counter 4.
11	aZ3g7wGqqru7AAA=	69 9d e0 ef 01 aa aa bb bb 00 00	NFC check-in (0x01): uid AA AA BB BB.
11	aZ3hCAKqqru7AAA=	69 9d e1 08 02 aa aa bb bb 00 00	NFC check-out (0x02):
11	aZ3hKQMCqqq7uwA=	69 9d e1 29 03 02 aa aa bb bb 00	NFC vote (0x03): button 2, card AA AA BB BB.
20	aZ3YuRIAAAABAAA=	69 9d d8 b9 12 00 00 00 01 00 00	Counter sync (0x12): button 0, counter 1.
20	aZ3ZwBAM1wAAAAA=	69 9d d9 c0 10 0c d7 00 00 00 00	Battery (0x10): battery_mv = 0x0CD7 (3287 mV), percent 0, flags 0.
20	<i>(construct)</i>	68 68 00 00 13 00 00 00 00 00 00	Snapshot (0x13): example ts 0x68680000, last_cleaned 0, tz offset 0.
21	<i>(construct)</i>	14 01 03 01 01 04 01 01 00 00 00	Version (0x14): FW 1.3.1 , HW 1.4.1 , epd_enabled=1 ; trailing three bytes reserved.

Decoding reference: Most events: [0..3] = ts BE, [4] = evt. **Exception**
— **0x14 on FPort 21:** [0]=0x14, [1..3] fw, [4..6] hw, [7] EPD flag,
[8..10]=0.